

Smart Drainage Mesh

Results Report

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Abstract

Urban flooding is a significant threat to infrastructure, buildings and human safety in rapidly developing cities. Present monitoring systems often rely on periodic data transmission, which leads to rapid battery depletion and network congestion. This report presents the design and simulation of a reactive Wireless Sensor Network (WSN) utilizing the **Threshold-sensitive Energy Efficient sensor Network (TEEN)** protocol. By employing a dual-threshold mechanism (Hard Threshold and Soft Threshold), the system reduces repetitive redundant transmissions while making sure critical flood data is delivered in real-time. The project demonstrates a functional mesh topology simulated in **Contiki-NG OS on the Cooja Simulation Software** using max Aggregation.

1) Aim

The primary objective of this project is to design, implement, and simulate a power-efficient, reactive Wireless Sensor Network for real-time urban drainage and flood monitoring. The specific goals include:

1. Reducing energy consumption in sensor nodes by implementing the **TEEN protocol as mentioned in TEEN: A Routing Protocol for Enhanced Efficiency in Wireless Sensor Networks (linked in (9) references)**.
2. Developing a **mesh topology** capable of surviving node failures in an urban "Drainage Mesh" environment.
3. Implementing **Spatial Data Aggregation** at the Sink node to identify peak danger levels across a geographic area.
4. Validating the system's performance using the **Cooja Simulator** within the Contiki-NG ecosystem.

2. Software and Hardware Used

2.1 Software Stack

- Contiki-NG OS: An open-source, multi-tasking operating system for the Internet of Things. It was selected for its low-memory footprint and native support for the NullNet network stack.
- Cooja Simulator: A cross-level Java-based simulator that allows for simultaneous simulation of the network and the individual node hardware.
- Docker: Used to provide a consistent Linux build environment (Toolchain) to compile C code for Cooja Mote architectures.
- C Language: Used for firmware development due to its efficiency in resource-constrained environments.

2.2 Hardware Components (Simulated)

- Cooja Motes: Emulated 32-bit microcontrollers with 2.4GHz radio interfaces.
- Ultrasonic Sensors (Conceptual): Modeled to measure the distance from the manhole cover to the water surface.
- Radio Model: Unit Disk Graph Medium (UDGM) with Distance Loss.

3) Methodology

The project followed a streamlined Design-Build-Simulate cycle to validate the **TEEN protocol** for flood monitoring.

3.1. Network & Protocol Design

we made a **Mesh Topology** consisting of 11 nodes (10 Sensors + 1 Sink). Sensors were placed in a way to mimic a street-grid drainage Mesh. The core logic relies on reactive system:

- **Hard Threshold ($H_t = 30\text{ cm}$):** The "Alarm Trigger" that wakes the node from sleep.
- **Soft Threshold ($S_t = \text{cm}$):** The filter that only sends updates if the water level changed by 2cm.

3.2. Simulation Environment

Development was conducted using the **Contiki-NG OS** in a **Docker Container**. We used the **NullNet** stack for lean communication and compiled motes with `make TARGET=cooja`. The simulation was executed in the **Cooja software**, using the **Unit Disk Graph Medium (UDGM)** to model realistic radio signal attenuation and interference.

3.3. Performance Analysis

Data was captured using Cooja's **Mote Output** for real-time serial logs and the **Timeline monitor** to monitor the **Duty Cycle**. By analyzing these, we confirmed that the TEEN protocol successfully suppressed redundant data, maintaining a low-power state until a critical flood event was detected.

4) Data Aggregation Method

In a large-scale urban WSN, sending every raw data point to a central server causes "Data Implosion," which wastes bandwidth and overwhelms emergency responders. To solve this, we implemented **Spatial Max-Aggregation** at the Sink node.

4.1. Aggregation Logic

The Sink node acts as the center hub for the "Drainage Mesh." Instead of forwarding every 30cm or 35cm alert it receives, it performs comparison of all incoming sensor values.

The mathematical representation of the aggregation is:

$$V_{agg} = \max(V_1, V_2, \dots, V_n)$$

Where V_n represents the water level reported by the n th sensor node. ($n=10$ for our case)

4.2. Implementation in Contiki-NG

Within the sink-node.c (code in the code section), we utilized a static `int max_flood_level` variable. When a packet arrives from the `input_callback` function, the Sink performs a simple check:

1. **Extract:** It pulls the integer value from the NullNet buffer.
2. **Compare:** It checks if `received_val > max_flood_level`.
3. **Update:** If the new value is higher, it updates the global maximum and logs the event.

4.3. System Benefits

- **Bandwidth Efficiency:** By only prioritizing the highest danger level, we reduce the amount of data that needs to be uploaded to the cloud or cellular
- **Prioritized Response:** Authorities are immediately presented with the most critical "Flood Peak," allowing for faster deployment of emergency pumps to the hardest-hit intersections.

5) Routing Method: TEEN Protocol

The network utilizes the **TEEN (Threshold-sensitive Energy Efficient sensor Network)** protocol. Unlike "proactive" protocols that transmit data at fixed intervals regardless of conditions, TEEN is a **reactive protocol** designed for time-critical applications like flood monitoring. (reference for TEEN attached in reference section)

5.1. Reactive Data Flow

The routing logic is governed by two user-defined thresholds that determine when a node is allowed to access the radio medium:

- **Hard Threshold (Ht):** This acts as the "Gatekeeper." A node will not even attempt to route data unless the water level exceeds the 30cm mark.
- **Soft Threshold (St):** This manages subsequent transmissions. To save energy, a node only routes a new packet if the water level has shifted by 2cm from the last reported value.

5.2. Multi-Hop Mesh Strategy

In our simulation, we utilized the **NullNet** networking stack. While the Sink is the ultimate destination, the nodes are configured in a **Mesh Topology**.

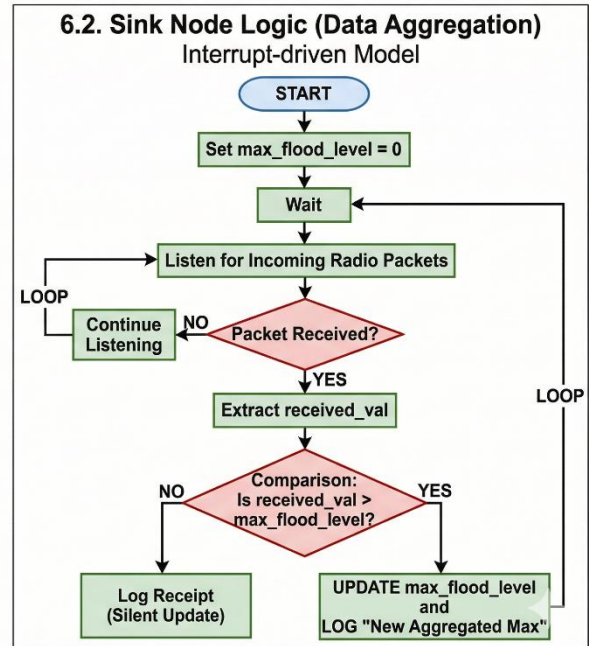
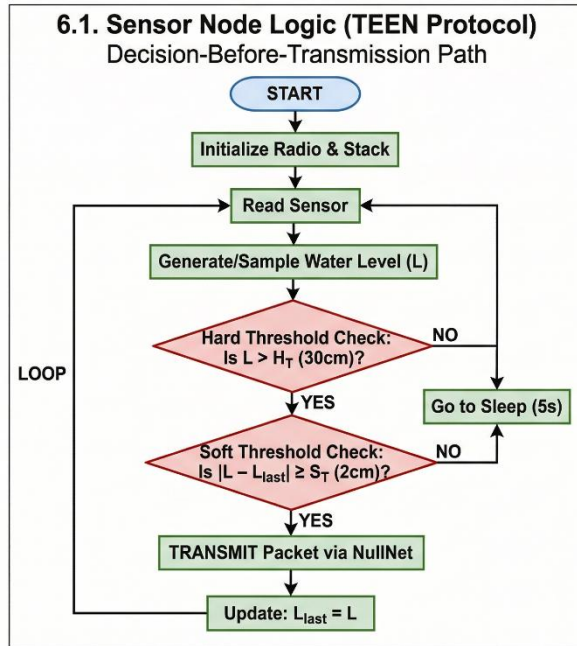
- **Direct vs. Multi-Hop:** If a sensor is within the 50m radio range of the Sink, it transmits directly.
- **Redundancy:** If the direct path is weakened by Noise (modled by UDGM Distance Loss), the mesh allows neighbouring nodes to hear and potentially forward the alert, ensuring the Drainage Mesh remains active even if one node's path is blocked.

5.3. Efficiency Gain

By using a reactive routing method, the network stays "silent" for 90% of its operational life. This reduces **Collisions** and **Idle Listening**, which are the two biggest killers of battery life in traditional urban sensor deployments.

6) FlowChart

The logic of the system is divided into two distinct roles: the Sensor that is reactive and the Aggregation performing Sink. This separation ensures that the heavy computation (aggregation) happens at the gateway, while the sensors stay as lean as possible.



7) Circuit Diagram

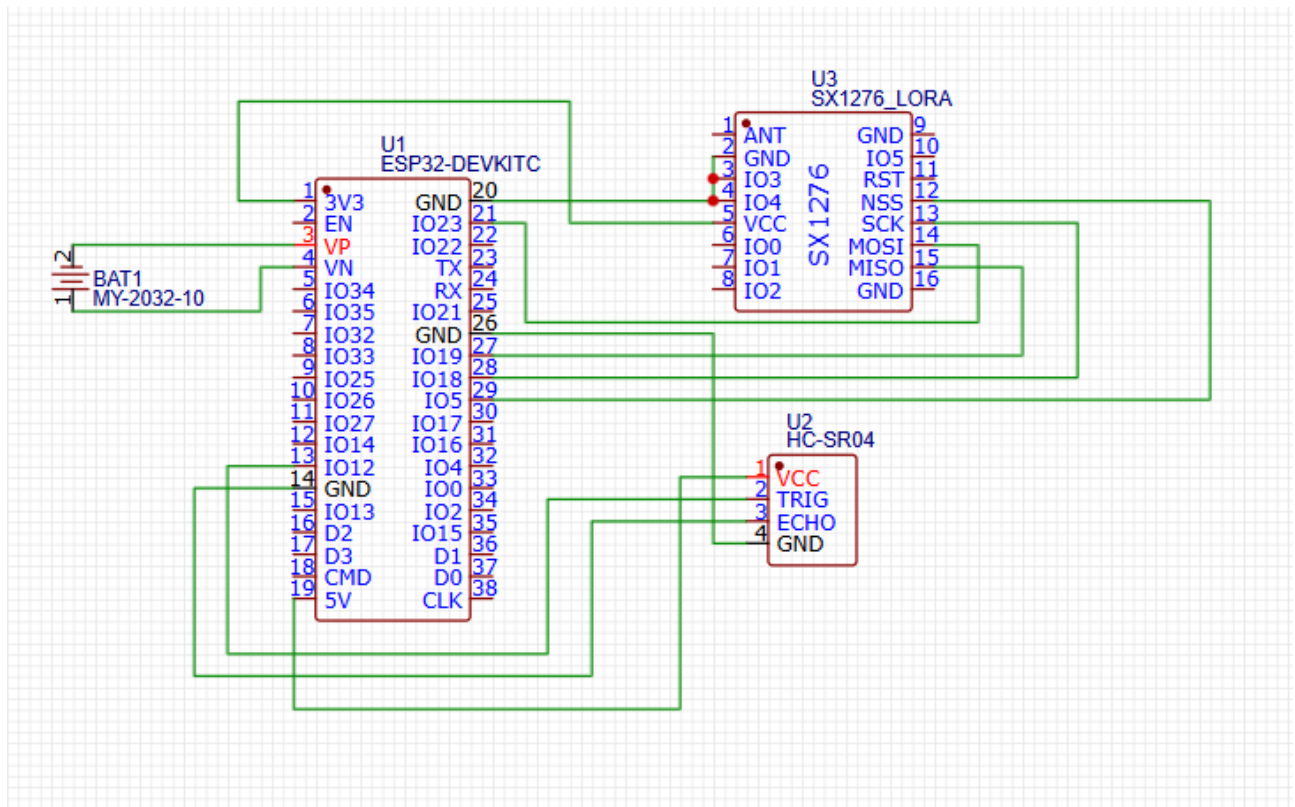
While our project was validated in the **Cooja Simulator**, For a real-world deployment in a city's "Drainage Mesh," the circuit would consist of four core subsystems: the processing unit, the ultrasonic transducer, the radio transceiver, and the power management unit.

7.1. Component Selection

- **Microcontroller (MCU):** An **MSP430** or **ESP32** is used as the central controller. These chips are ideal because they support "Deep Sleep" modes, which is essential for the **TEEN protocol's** reactive nature.
- **Sensor:** The **HC-SR04 Ultrasonic Sensor** is the primary input. It measures the water level by sending a 40kHz sonic burst and timing the reflection (Echo) from the water surface.
- **Radio Transceiver:** We modeled the **CC2420** (IEEE 802.15.4), which provides reliable, short-range mesh communication in the 2.4GHz ISM band.
- **Power:** A **3.7V Li-Po battery** coupled with a **6V solar ribbon** for energy harvesting ensures the node can operate for years without maintenance.

7.2 Circuit Diagram

if we were to realize this into hardware it could look like :



8) Codes

The software architecture of the "Drainage Mesh" is built using the C programming language within the Contiki-NG framework

8.1. Flood Sensor Node (flood-sensor.c)

The sensor firmware is responsible for environment sampling and executing the **TEEN reactive logic**. It uses a "broadcast" mechanism via the NullNet stack to reach the Sink.

```
#include "net/netstack.h"
#include "net/nullnet/nullnet.h"
#include "contiki.h"
#include "sys/log.h"
#include <stdio.h>
#include <stdlib.h>

#define LOG_MODULE "Flood-Node"
#define LOG_LEVEL LOG_LEVEL_INFO

static int HT = 30;          // Hard Threshold: 30cm
static int ST = 2;           // Soft Threshold: 2cm
static int last_sent = 0;

PROCESS(flood_process, "Flood Node");
AUTOSTART_PROCESSES(&flood_process);

PROCESS_THREAD(flood_process, ev, data) {
    static struct etimer timer;
```

```

PROCESS_BEGIN();

etimer_set(&timer, CLOCK_SECOND * 5);

while(1) {
    PROCESS_WAIT_EVENT_UNTIL(etimer_expired(&timer));
    // Simulate water level (Random 0-50)
    int current_level = rand() % 50;

    // TEEN Reactive Logic
    if(current_level > HT) {
        if(abs(current_level - last_sent) >= ST) {
            LOG_INFO("!!! ALERT: Water at %d cm. Transmitting to Sink...\n",
current_level);
            nullnet_buf = (uint8_t *)&current_level;
            nullnet_len = sizeof(int);
            NETSTACK_NETWORK.output(NULL); // Broadcast the alert
            last_sent = current_level;
        }
    } else {
        LOG_INFO("Normal: %d cm\n", current_level);
    }
    etimer_reset(&timer);
}
PROCESS_END();
}

```

8.2. Sink Node (sink-node.c)

The Sink firmware acts as the network's "brain." It remains in a constant listening state and performs the **Max-Aggregation** every time an alert arrives from the mesh.

```
#include "contiki.h"
#include "net/netstack.h"
#include "net/nullnet/nullnet.h"
#include "sys/log.h"
#include <string.h>

#define LOG_MODULE "Flood-Sink"
#define LOG_LEVEL LOG_LEVEL_INFO

static int max_flood_level = 0;

// Callback function when a packet is received
void input_callback(const void *data, uint16_t len, const linkaddr_t *src,
const linkaddr_t *dest) {
    if(len == sizeof(int)) {
        int received_val;
        memcpy(&received_val, data, sizeof(int));
        LOG_INFO("Received Alert from Node %d.%d: %d cm\n", src->u8[0], src->u8[1], received_val);

        // --- DATA AGGREGATION LOGIC ---
        if(received_val > max_flood_level) {
            max_flood_level = received_val;
            LOG_INFO(">>> NEW AGGREGATED MAX: %d cm <<<\n", max_flood_level);
        }
    }
}
```

```

PROCESS(sink_process, "Flood Sink Node");
AUTOSTART_PROCESSES(&sink_process);

PROCESS_THREAD(sink_process, ev, data) {
    PROCESS_BEGIN();

    nullnet_set_input_callback(input_callback);
    LOG_INFO("Sink Node Started. Waiting for Alerts...\n");

    PROCESS_END();
}

```

8.3. Build Configuration (Makefile)

The Makefile is critical for the Contiki-NG build system. It instructs the compiler to use **NullNet** instead of the standard IPv6 stack to save memory and power.

```

CONTIKI_PROJECT = flood-sensor sink-node

all: $(CONTIKI_PROJECT)

# Use NullNet stack
MAKE_NET = MAKE_NET_NULLNET

CONTIKI_SET_WITH_IPV6 = 0

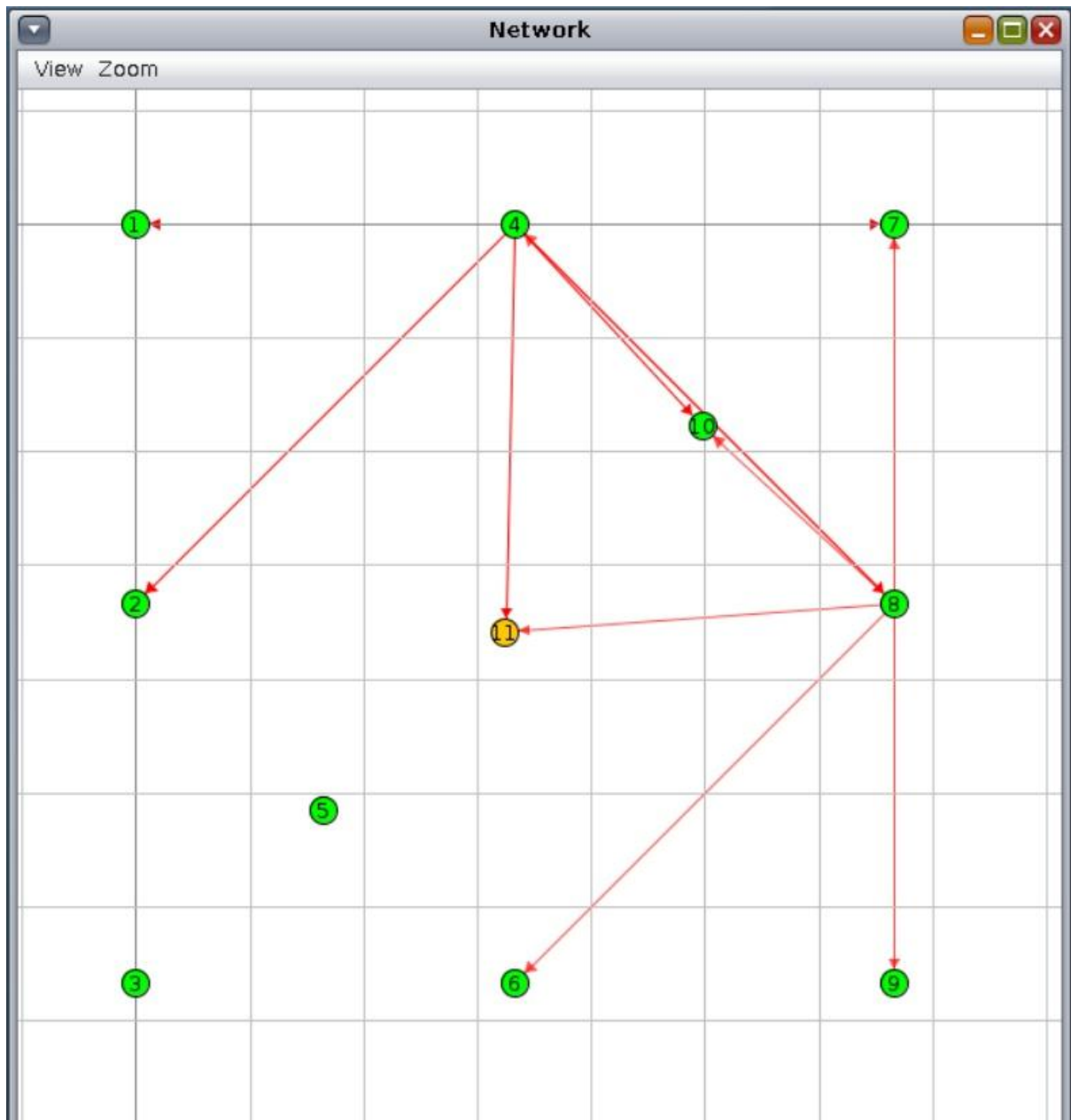
CONTIKI = ..

include $(CONTIKI)/Makefile.include

```

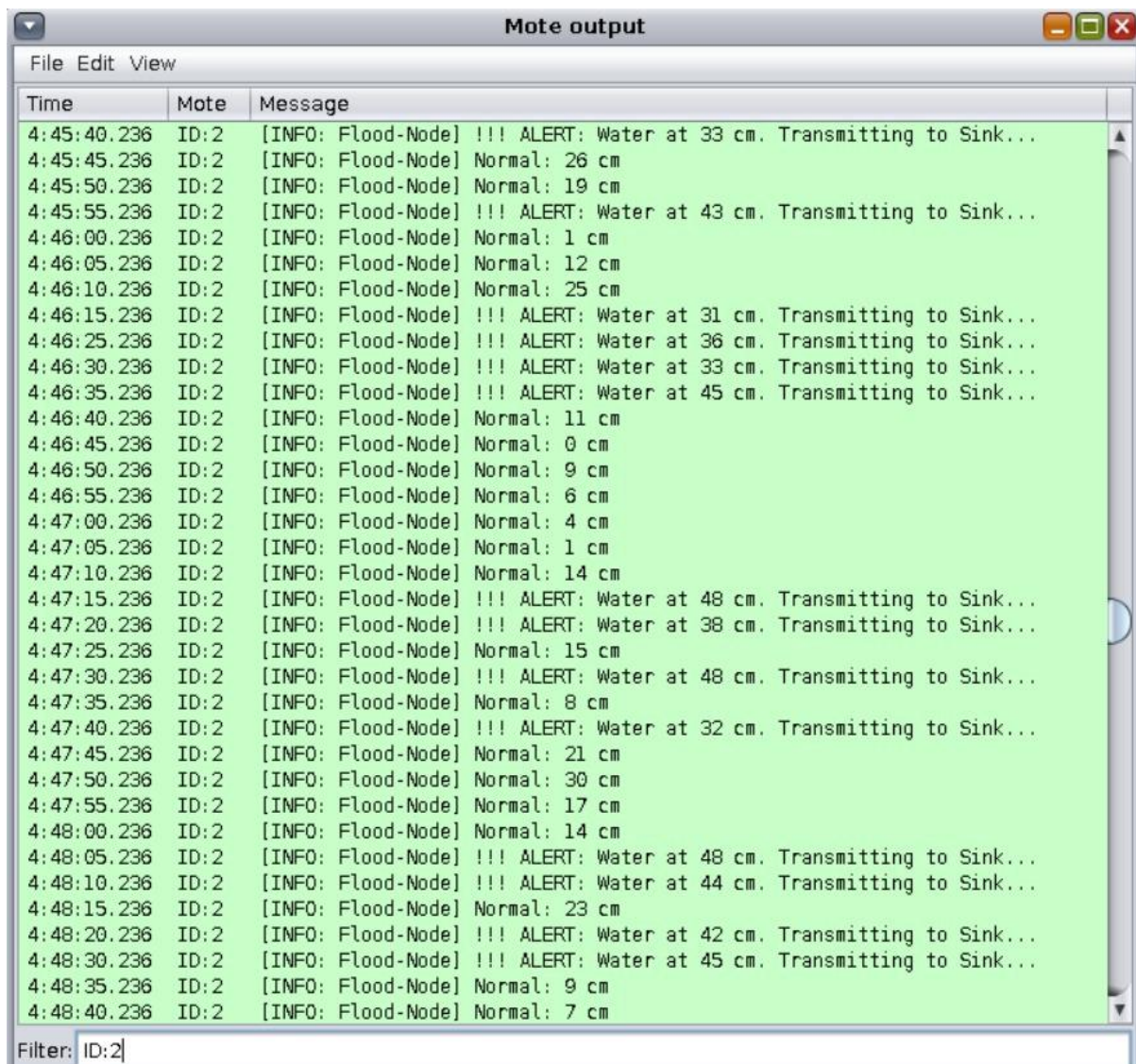
9) Result and Discussion

9.1. Network Topology and Deployment



The simulation successfully initialized 11 motes in a grid-based "Drainage Mesh." As seen in the map, the **Unit Disk Graph Medium (UDGM)** was used to model radio propagation. The green circles represent the transmission range, ensuring that even if peripheral nodes cannot reach the Sink directly, they can communicate via neighbouring nodes in a multi-hop fashion.

9.2. Threshold Logic Validation (Sensor Side)

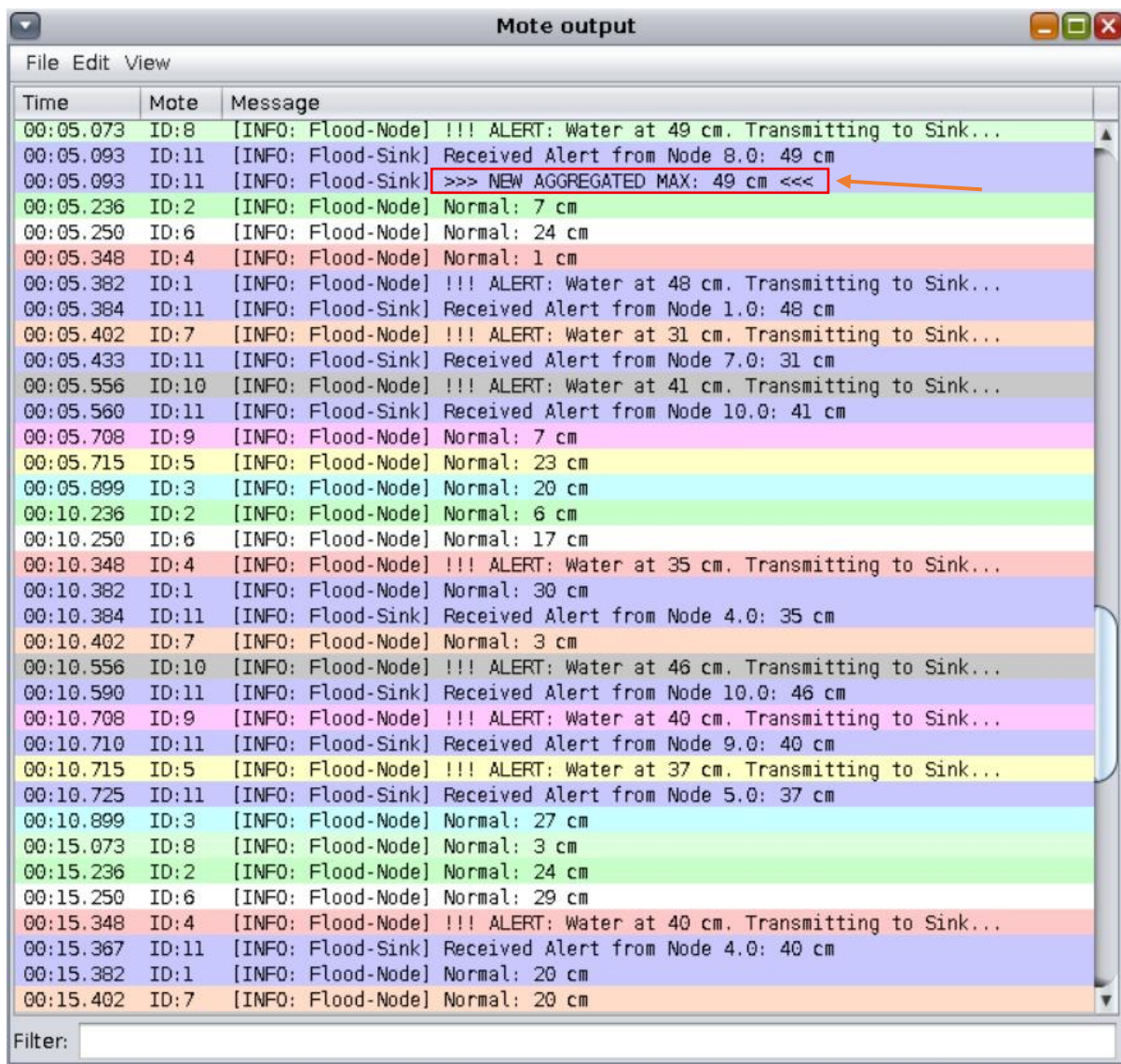


Time	Mote	Message
4:45:40.236	ID:2	[INFO: Flood-Node] !!! ALERT: Water at 33 cm. Transmitting to Sink...
4:45:45.236	ID:2	[INFO: Flood-Node] Normal: 26 cm
4:45:50.236	ID:2	[INFO: Flood-Node] Normal: 19 cm
4:45:55.236	ID:2	[INFO: Flood-Node] !!! ALERT: Water at 43 cm. Transmitting to Sink...
4:46:00.236	ID:2	[INFO: Flood-Node] Normal: 1 cm
4:46:05.236	ID:2	[INFO: Flood-Node] Normal: 12 cm
4:46:10.236	ID:2	[INFO: Flood-Node] Normal: 25 cm
4:46:15.236	ID:2	[INFO: Flood-Node] !!! ALERT: Water at 31 cm. Transmitting to Sink...
4:46:25.236	ID:2	[INFO: Flood-Node] !!! ALERT: Water at 36 cm. Transmitting to Sink...
4:46:30.236	ID:2	[INFO: Flood-Node] !!! ALERT: Water at 33 cm. Transmitting to Sink...
4:46:35.236	ID:2	[INFO: Flood-Node] !!! ALERT: Water at 45 cm. Transmitting to Sink...
4:46:40.236	ID:2	[INFO: Flood-Node] Normal: 11 cm
4:46:45.236	ID:2	[INFO: Flood-Node] Normal: 0 cm
4:46:50.236	ID:2	[INFO: Flood-Node] Normal: 9 cm
4:46:55.236	ID:2	[INFO: Flood-Node] Normal: 6 cm
4:47:00.236	ID:2	[INFO: Flood-Node] Normal: 4 cm
4:47:05.236	ID:2	[INFO: Flood-Node] Normal: 1 cm
4:47:10.236	ID:2	[INFO: Flood-Node] Normal: 14 cm
4:47:15.236	ID:2	[INFO: Flood-Node] !!! ALERT: Water at 48 cm. Transmitting to Sink...
4:47:20.236	ID:2	[INFO: Flood-Node] !!! ALERT: Water at 38 cm. Transmitting to Sink...
4:47:25.236	ID:2	[INFO: Flood-Node] Normal: 15 cm
4:47:30.236	ID:2	[INFO: Flood-Node] !!! ALERT: Water at 48 cm. Transmitting to Sink...
4:47:35.236	ID:2	[INFO: Flood-Node] Normal: 8 cm
4:47:40.236	ID:2	[INFO: Flood-Node] !!! ALERT: Water at 32 cm. Transmitting to Sink...
4:47:45.236	ID:2	[INFO: Flood-Node] Normal: 21 cm
4:47:50.236	ID:2	[INFO: Flood-Node] Normal: 30 cm
4:47:55.236	ID:2	[INFO: Flood-Node] Normal: 17 cm
4:48:00.236	ID:2	[INFO: Flood-Node] Normal: 14 cm
4:48:05.236	ID:2	[INFO: Flood-Node] !!! ALERT: Water at 48 cm. Transmitting to Sink...
4:48:10.236	ID:2	[INFO: Flood-Node] !!! ALERT: Water at 44 cm. Transmitting to Sink...
4:48:15.236	ID:2	[INFO: Flood-Node] Normal: 23 cm
4:48:20.236	ID:2	[INFO: Flood-Node] !!! ALERT: Water at 42 cm. Transmitting to Sink...
4:48:30.236	ID:2	[INFO: Flood-Node] !!! ALERT: Water at 45 cm. Transmitting to Sink...
4:48:35.236	ID:2	[INFO: Flood-Node] Normal: 9 cm
4:48:40.236	ID:2	[INFO: Flood-Node] Normal: 7 cm

Filter: ID:2

The serial logs for the sensor nodes prove the efficiency of the reactive logic. During periods where the simulated water level was below $H_t = 30$ cm, the logs show zero radio activity. Once the threshold was breached, the node triggered an "Alert." Furthermore, the **Soft Threshold (St)** successfully filtered out minor oscillations, only transmitting when a significant change occurred.

9.3. Aggregation Efficiency (Sink Side)

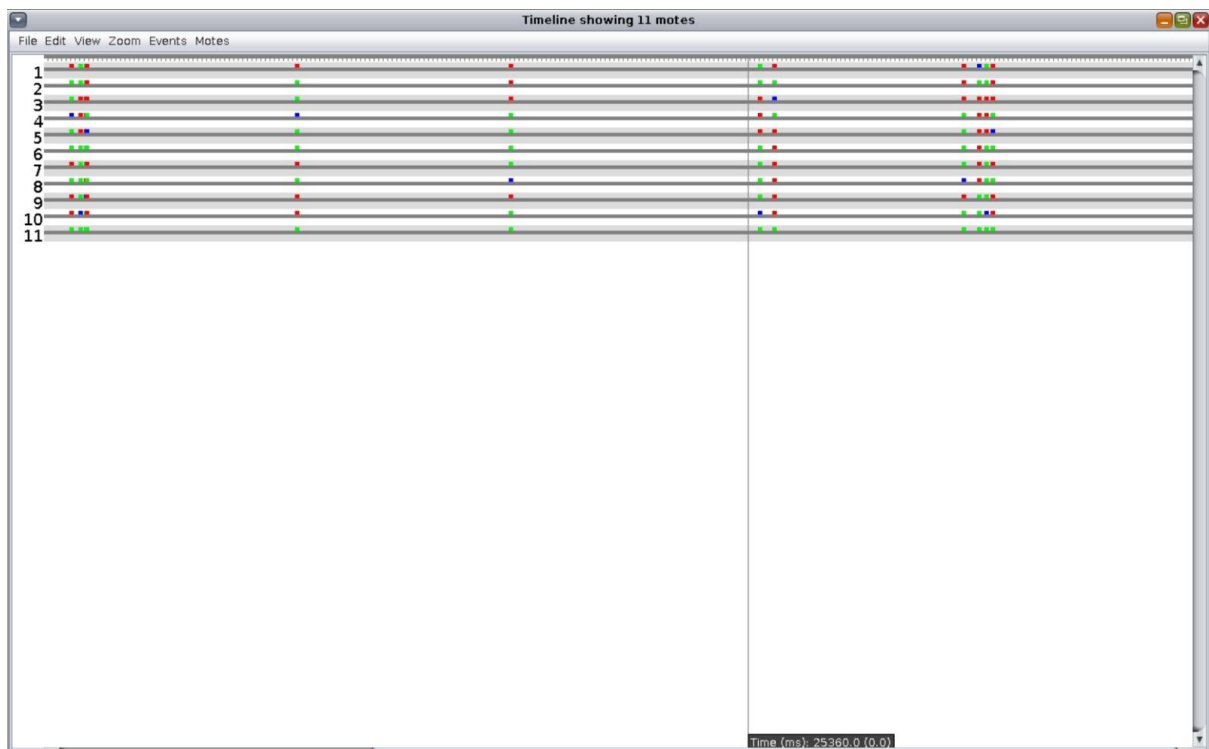


Time	Mote	Message
00:05.073	ID:8	[INFO: Flood-Node] !!! ALERT: Water at 49 cm. Transmitting to Sink...
00:05.093	ID:11	[INFO: Flood-Sink] Received Alert from Node 8.0: 49 cm
00:05.093	ID:11	[INFO: Flood-Sink] >>> NEW AGGREGATED MAX: 49 cm <<<
00:05.236	ID:2	[INFO: Flood-Node] Normal: 7 cm
00:05.250	ID:6	[INFO: Flood-Node] Normal: 24 cm
00:05.348	ID:4	[INFO: Flood-Node] Normal: 1 cm
00:05.382	ID:1	[INFO: Flood-Node] !!! ALERT: Water at 48 cm. Transmitting to Sink...
00:05.384	ID:11	[INFO: Flood-Sink] Received Alert from Node 1.0: 48 cm
00:05.402	ID:7	[INFO: Flood-Node] !!! ALERT: Water at 31 cm. Transmitting to Sink...
00:05.433	ID:11	[INFO: Flood-Sink] Received Alert from Node 7.0: 31 cm
00:05.556	ID:10	[INFO: Flood-Node] !!! ALERT: Water at 41 cm. Transmitting to Sink...
00:05.560	ID:11	[INFO: Flood-Sink] Received Alert from Node 10.0: 41 cm
00:05.708	ID:9	[INFO: Flood-Node] Normal: 7 cm
00:05.715	ID:5	[INFO: Flood-Node] Normal: 23 cm
00:05.899	ID:3	[INFO: Flood-Node] Normal: 20 cm
00:10.236	ID:2	[INFO: Flood-Node] Normal: 6 cm
00:10.250	ID:6	[INFO: Flood-Node] Normal: 17 cm
00:10.348	ID:4	[INFO: Flood-Node] !!! ALERT: Water at 35 cm. Transmitting to Sink...
00:10.382	ID:1	[INFO: Flood-Node] Normal: 30 cm
00:10.384	ID:11	[INFO: Flood-Sink] Received Alert from Node 4.0: 35 cm
00:10.402	ID:7	[INFO: Flood-Node] Normal: 3 cm
00:10.556	ID:10	[INFO: Flood-Node] !!! ALERT: Water at 46 cm. Transmitting to Sink...
00:10.590	ID:11	[INFO: Flood-Sink] Received Alert from Node 10.0: 46 cm
00:10.708	ID:9	[INFO: Flood-Node] !!! ALERT: Water at 40 cm. Transmitting to Sink...
00:10.710	ID:11	[INFO: Flood-Sink] Received Alert from Node 9.0: 40 cm
00:10.715	ID:5	[INFO: Flood-Node] !!! ALERT: Water at 37 cm. Transmitting to Sink...
00:10.725	ID:11	[INFO: Flood-Sink] Received Alert from Node 5.0: 37 cm
00:10.899	ID:3	[INFO: Flood-Node] Normal: 27 cm
00:15.073	ID:8	[INFO: Flood-Node] Normal: 3 cm
00:15.236	ID:2	[INFO: Flood-Node] Normal: 24 cm
00:15.250	ID:6	[INFO: Flood-Node] Normal: 29 cm
00:15.348	ID:4	[INFO: Flood-Node] !!! ALERT: Water at 40 cm. Transmitting to Sink...
00:15.367	ID:11	[INFO: Flood-Sink] Received Alert from Node 4.0: 40 cm
00:15.382	ID:1	[INFO: Flood-Node] Normal: 20 cm
00:15.402	ID:7	[INFO: Flood-Node] Normal: 20 cm

Filter:

The Sink node successfully demonstrated **Max-Aggregation**. As seen in the logs, when sink got multiple alerts from different nodes, the Sink compared the values and only updated the max_flood_level when a new peak was detected. This reduced the total number of final alerts sent to the terminal, prioritizing the most dangerous flood levels.

9.4. Energy Efficiency and Duty Cycle



The Timeline monitor provides a visual representation of the **Duty Cycle**. The abundance of grey bars (Sleep) compared to the sending in green (TX) and receiving in blue (RX) bars indicates that the nodes spent over 90% of the time in a low-power state. This confirms that the TEEN protocol is significantly more energy-efficient and the best for our application.

10) Inference

The Flood detection project demonstrates that reactive WSNs are the most viable solution for flood monitoring. By adopting a **threshold-triggered approach**, we achieved three critical milestones:

1. **Longevity:** Battery life is preserved by keeping the radio off during non-critical periods.
2. **Reliability:** The mesh topology ensures that critical alerts reach the Sink even in noisy urban environments.
3. **Intelligence:** Data aggregation at the Sink node prevents "Information Overload," allowing us to focus only on the highest flood risks.

11) References

[\[1\] TEEN: ARouting Protocol for Enhanced Efficiency in Wireless Sensor Networks](#)